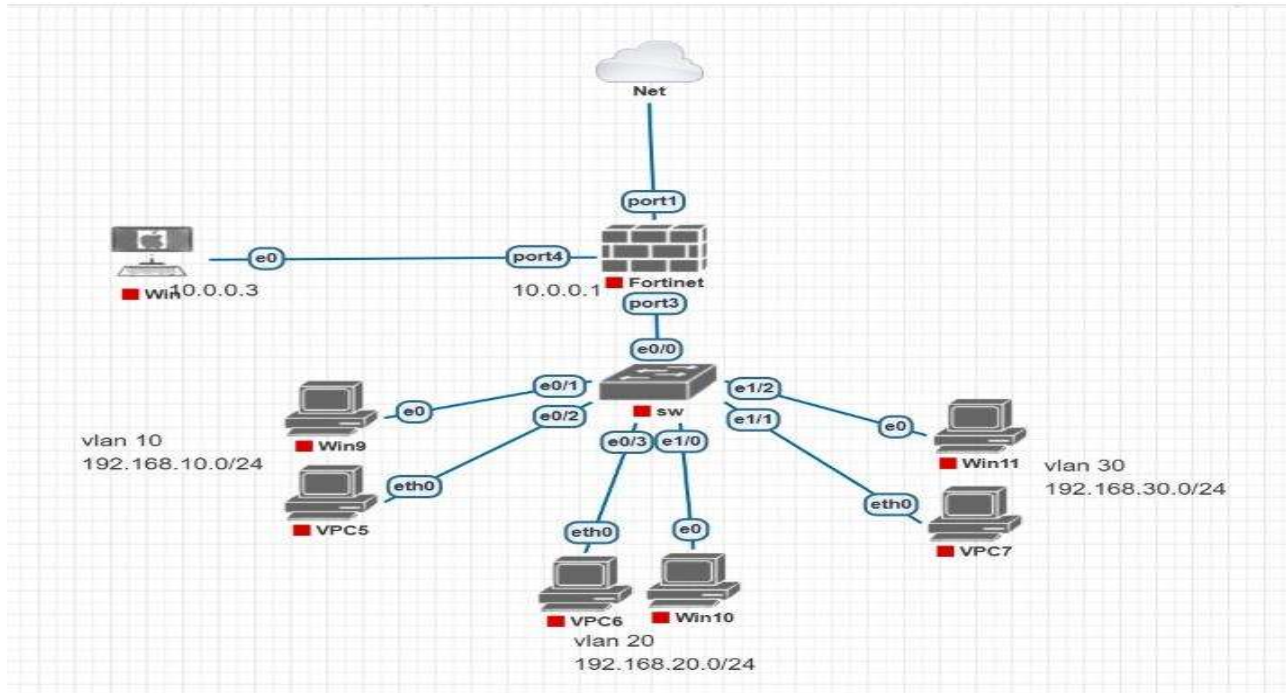




Low Level Design (LLD)

1. Overview

This design represents a virtual lab network consisting of:

- A **Fortinet Firewall**
- A Layer 2 switch
- Three VLANs:
 - VLAN 10: Network 192.168.10.0/24
 - VLAN 20: Network 192.168.20.0/24
 - VLAN 30: Network 192.168.30.0/24
- External connectivity through the 10.0.0.0/24 network

2. Devices and Interfaces

Device	Interface	IP Address	Notes
Fortinet	port1	DHCP/ISP	External Internet Access
Fortinet	port3	Trunk	Connected to the switch (core)
Fortinet	port4	10.0.0.1	Connected to a management PC
PC (Win)	e0	10.0.0.3	For direct management
Switch	e0/0-1-3	Access VLAN 10	Hosts: Win9, VPC5
Switch	e1/0	Access VLAN 20	Hosts: Win10, VPC6
Switch	e1/1-2	Access VLAN 30	Hosts: Win11, VPC7

3. VLAN Configuration

VLAN ID	Subnet	Hosts	Gateway
10	192.168.10.0/24	Win9, VPC5	192.168.10.1
20	192.168.20.0/24	Win10, VPC6	192.168.20.1
30	192.168.30.0/24	Win11, VPC7	192.168.30.1

All gateways are configured on the Fortinet. The connection between the Fortinet and the switch is a trunk link.

4. Switch Configuration

- All host-facing ports: Access mode with assigned VLANs.
- Port connected to Fortinet: Trunk mode, allowing VLANs 10, 20, 30.

5. Routing & NAT (on Fortinet)

- **Inter-VLAN Routing:** Enabled on the Fortinet using SVIs (interfaces for each VLAN).
- **NAT:** From internal VLANs to the internet via port1.
- Static Route:
 - 0.0.0.0/0 → via port1 (default route to internet)

6. Firewall Rules

- Allow:
 - VLANs → Internet
 - Inter-VLAN traffic (optional, depending on need)
 - Management access to Fortinet from 10.0.0.3
- Deny any unknown/incoming traffic from outside

7. Services

- **DHCP:**
 - Can be configured on Fortinet for each VLAN or use static IPs.
 - **DNS:**
 - Configured on Fortinet or forwarded to public DNS (e.g., 8.8.8.8)
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8. Optional Enhancements

- Enable **Ping/ICMP** for troubleshooting
- ACLs between VLANs for access control
- Logging & Monitoring enabled on Fortinet

9. Fortinet Firewall Policy Table (Explicit Rules)

Source	Destination	Service	Action	NAT	Notes
VLAN 10	Internet	ALL	Allow	Yes	Basic Internet Access
VLAN 20	Internet	ALL	Allow	Yes	Basic Internet Access
VLAN 30	Internet	ALL	Allow	Yes	Basic Internet Access
VLAN 10	VLAN 20	Specific	Allow	No	Optional Inter-VLAN
VLAN 10	VLAN 30	Specific	Allow	No	Optional Inter-VLAN
VLAN 20	VLAN 10	Specific	Allow	No	Optional Inter-VLAN
VLAN 20	VLAN 30	Specific	Allow	No	Optional Inter-VLAN
VLAN 30	VLAN 10	Specific	Allow	No	Optional Inter-VLAN
VLAN 30	VLAN 20	Specific	Allow	No	Optional Inter-VLAN
WAN	Any	Any	Deny	No	Default Deny All